Importance Sampling with Real-Valued Weights under Differential Privacy

Wakhee Ahmed

a1985586

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School of Mathematical Sciences,
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Project Supervisor: **Dr Joshua Bon**

Abstract

A data holder can release a weighted sample of parameter values that represents a Bayesian posterior, and analysts can then compute any posterior quantity from it at no further privacy cost. Such an importance sample can be privatised by adding zero-mean noise to the weights, in which case some noisy weights are negative. Larger noise gives stronger privacy but lower utility, and the aim of this work is to find the best balance between the two. This report studies the unnormalised importance-sampling estimator under that scheme. The noisy estimator is unbiased at every number of particles N, so its mean squared error equals its variance, and that variance is derived in terms of the variance of the estimator without noise. Under Laplace noise calibrated for ε-differential privacy with basic composition, the variance is the sum of a Monte Carlo term that decreases like $1/N$ and a privacy term that grows like N, so at a fixed budget it has a positive lower bound that holds for every number of particles. A simulation on a normal model with ten records agrees with this formula, and an attack that tries to recover one record from the release does better than a simple reference guess only at budgets above about 10^5.

1 Introduction

Statistical inference frequently uses sensitive personal data, such as patient medical records, and its results are often shared outside the institution that holds the data. Differential privacy (DP) limits what such a release can reveal: replacing any one person's record changes the probability of any output by at most a fixed factor [5, 6]. Each release spends part of a privacy budget, and the costs add up under composition [6].

In Bayesian inference the quantities of interest are posterior expectations, and an analyst typically wishes to calculate several such quantities. Releasing each quantity through its own private mechanism spends part of the privacy budget every time, so the budget is quickly exhausted. An alternative is to release a weighted sample that represents the posterior once. Anything computed from that sample afterwards is post-processing and costs no further privacy [6].

Example 1 (An HIV drug trial)**.** A trial of a new HIV drug runs at two hospitals. For each patient the trial records their age, their hospital, whether they received the drug, and a blood test result that shows how far the infection has progressed, such as the CD4 count [12]. Researchers outside the trial want several quantities from these records: the average change in the test result due to the drug, the uncertainty in that change, the probability that the change is positive, and the result a new patient should expect.

The records cannot be released as they are. Every participant has HIV, so any record that can be traced to a person reveals their HIV status together with their age and test results. A separate private release for each quantity would spend part of the privacy budget on every one, and the researchers may have further questions later. A single private release that represents the posterior allows all of these quantities, and later ones, to be computed at no further privacy cost to the patients.

Importance sampling produces such a sample. Parameter values, called particles, are drawn from a proposal distribution, and each is weighted in proportion to how much more probable it is under the posterior than under the proposal [8]. The sample can be privatised by adding independent zero-mean noise to the weights before release [2], in which case some noisy weights are negative. Chopin and Papaspiliopoulos [4] observe that random weights need not be non-negative for importance sampling to be valid, as long as each has the right conditional expectation. In the pri-

vacy setting negative weights arise by design, and the privacy budget determines the scale of the noise that produces them.

There is an inherent trade-off between privacy and utility. More noise gives stronger privacy and less accurate estimates, and Karmitsa et al. describe balancing the two as a central problem in releasing sensitive data [9]. Importance sampling introduces a third quantity, the number of particles N, which determines the computational cost. The problem is therefore a three-way trade-off between privacy, utility and computation, and the question addressed in this report is whether an optimal trade-off can be found at a fixed privacy budget.

This report asks how DP noise affects the variance and consistency of the unnormalised importance-sampling estimator when the noise is calibrated to protect all N noisy weights together. The report makes three contributions.

1. The noisy estimator is proved unbiased at every N, and its variance is given in terms of the variance of the estimator without noise (Sections 3.2 and 3.3).
2. The Laplace noise scale required under basic composition is derived (Section 3.4). Under this scale the variance is a Monte Carlo term decreasing like $1/N$ plus a privacy term increasing like N, so at a fixed budget it is minimised at a finite N and cannot be made arbitrarily small (Section 3.5).
3. A simulation on a normal model checks the variance formula numerically, and measures how well an adversary can recover one record from the release over a range of budgets (Section 4).

2 Background

2.1 Bayesian inference and importance sampling

A statistical model describes how the observed data are generated. It treats the data x as a draw from a distribution $L(x \mid \theta)$ that depends on an unknown parameter θ [12]. There can be several parameters, collected in a vector θ of any finite dimension [12].

Example 1 (continued)**.** The data x are the trial's records, one per patient. The simplest model has one parameter, the drug's effect on the test result, and models each patient's test result as that effect plus random error. For example, a regression model of the trial [7] could have five parameters, collected in the vector $\theta = (\theta_1, \ldots, \theta_5)$.

1. The baseline θ_1 is the average test result of an untreated patient.
2. The treatment effect θ_2 is the average change in the test result due to the drug.
3. The age effect θ_3 is the change in the test result for each extra year of age.
4. The hospital effect θ_4 is the difference in test results between the two hospitals.
5. The spread θ_5 is the variance of patients' test results around what the other four parameters predict.

In probability, the parameter is given and the model specifies the distribution of the data. Statistical inference proceeds in the opposite direction, from the observed data to the parameter [12]. Robert treats this reversal as the central aim of statistical analysis: the observations are the effects, the parameters are the causes, and inference works back from one to the other [12].

The likelihood is the density of the data, regarded as a function of θ with the observed x held fixed [12]. It measures how probable the observed data are at each value of θ.

A Bayesian model also assigns a distribution to θ itself, the prior density $p(\theta)$, which describes the values of θ that are plausible before the data are observed [12]. Bayes' rule then carries out this reversal and gives the posterior density

$$p(\theta \mid x) = \frac{p(\theta)L(x \mid \theta)}{Z}, \qquad Z = \int p(\theta)L(x \mid \theta)\, d\theta, \tag{1}$$

which describes the values of θ that are plausible after the data are observed. The number Z is the normalising constant. It is the density of the data averaged over the prior, and it is related to the data's support for the model in question.

Many quantities of interest can be written as a posterior expectation $\mathrm{E}(f(\theta) \mid x)$ for a test function f, and different choices of f give different quantities. When θ is a vector, f is still real-valued, so it may depend on a single component of θ.

Example 1 (continued)**.** Each quantity the researchers want is one choice of f, and the first three depend only on the treatment effect θ_2.

1. Taking $f(\theta) = \theta_2$ gives the posterior mean of the treatment effect, the drug's average effect on the test result given the data.
2. Taking $f(\theta) = \theta_2^2$ gives its posterior second moment. Together with the mean it gives the posterior variance, $\mathrm{var}(\theta_2 \mid x) = \mathrm{E}(\theta_2^2 \mid x) - \mathrm{E}(\theta_2 \mid x)^2$, which measures the uncertainty in the effect.
3. Taking $f(\theta)$ equal to one when $\theta_2 > 0$ and zero otherwise gives the posterior probability that the effect is positive.
4. Taking $f(\theta) = L(\tilde{x} \mid \theta)$ for a new patient's record $\tilde{x}$ gives the posterior predictive density of that record [7], which describes the result a new patient should expect.

The main difficulty is computing Z, which is an integral over the whole parameter space, with one dimension for each parameter. A closed form exists only in special cases, such as conjugate models, where the posterior belongs to the same family as the prior. For more complex models a conjugate form is rarely available [7].

Numerical integration does not solve this in general. Deterministic rules replace the integral with a weighted sum over chosen points, and as the dimension grows it becomes difficult to place those points effectively [7]. A grid with 100 points per parameter requires $100^5 = 10^{10}$ evaluations for the five parameters of Example 1, and each evaluation of the likelihood uses every record in the dataset. As a rule of thumb, Robert advises against most standard integration methods above four dimensions, and this breakdown is known as the curse of dimensionality [12].

Consequently, these expectations usually cannot be computed directly. The product $p(\theta)L(x \mid \theta)$ can be evaluated at any single θ, which allows candidate values to be compared, but this does not provide a way to draw exact samples from the posterior.

Importance sampling rewrites an integral with respect to one density as an integral with respect to another density that is easy to sample from

[4]. Draw particles $\theta_1, \ldots, \theta_N$ independently and identically distributed (i.i.d.) from a proposal density q that is positive wherever $p(\theta)L(x \mid \theta)f(\theta)$ is non-zero, and define the weight

$$w(\theta, x) = \frac{p(\theta)L(x \mid \theta)}{q(\theta)}. \tag{2}$$

Then

$$F = \int f(\theta)p(\theta)L(x \mid \theta)\, d\theta = \mathrm{E}_q\left(w(\theta, x)f(\theta)\right), \tag{3}$$

and the average

$$\frac{1}{N}\sum_{i=1}^{N} w(\theta_i, x)f(\theta_i) \tag{4}$$

is an unbiased estimator of F with variance $\mathrm{var}_q(wf)/N$ [4]. It has the form of what Chopin and Papaspiliopoulos call the normalised estimator, which assumes $\mathrm{E}_q(w) = 1$, while here $\mathrm{E}_q(w) = Z$. They write q for the target density and m for the proposal, whereas this report uses q for the proposal.

The weight can be viewed as a correction factor. Rearranging (1) and putting it into (2) gives

$$w(\theta, x) = Z\,\frac{p(\theta \mid x)}{q(\theta)},$$

so up to the constant Z it measures how much more probable a particle is under the posterior than under the proposal.

The variance of (4) decreases like $1/N$, so in ordinary Monte Carlo the particle count N acts purely as a computational budget, and more particles always give a more accurate estimate. Section 3 shows how privacy changes this.

An estimator is unbiased if its expectation equals its target at every N, and consistent if it converges in probability to its target as $N \to \infty$. By Chebyshev's inequality [1], an unbiased estimator whose variance goes to zero is consistent.

This report uses the unnormalised estimator (4). The estimator is called unnormalised because the weights are used as they are, without rescaling them to sum to one, so the average divides by N. By (1), its target is $F = Z\ \mathrm{E}(f(\theta) \mid x)$, the posterior expectation multiplied by the constant Z.

This report takes the proposal to be the prior, $q = p$, for simplicity. Particles are drawn directly from the prior, the prior then cancels in (2), and the weight reduces to the likelihood, $w(\theta, x) = L(x \mid \theta)$.

2.2 Differential privacy

Differential privacy compares the output distributions of an algorithm on two datasets that differ only in one person's record. Two datasets x and x' are neighbours, written $x \sim x'$, if they differ in one record. A randomised algorithm M is ε-differentially private if, for every pair of neighbours and every set S of possible outputs,

$$\Pr(M(x) \in S) \leq e^{\varepsilon} \Pr(M(x') \in S) \tag{5}$$

[6]. Dwork and Roth define neighbours by adding or removing a record. This report instead defines neighbours by replacing one record, and Appendix A verifies the one privacy result used here directly under that definition.

Example 1 (continued)**.** A neighbour of the trial's records is the same set of records with one patient's record replaced by another person's. Definition (5) says the release has almost the same distribution under both, so someone who sees it cannot tell with confidence which of the two records was used [6]. In particular, the release reveals little about whether a given person took part in the trial, and hence whether they have HIV.

The parameter ε is the privacy budget. When ε is small, e^{ε} is close to one and the distribution of the output changes little when one person's record changes, so the release reveals little about any individual. A large ε allows the two distributions to differ substantially, and an observer of the release may be able to determine which of the two datasets produced it. The right value depends on the application. Karmitsa et al. class $\varepsilon \leq 1$ as strong privacy, $1 < \varepsilon \leq 10$ as moderate and anything larger as weak, and values below 1 are usual for simple statistics such as a mean [9]. A smaller ε requires more noise, and Section 3 quantifies the resulting loss of accuracy. Definition (5) is pure differential privacy [6], and this report uses it with Laplace noise throughout.

A query is a function of the dataset. The sensitivity of a real-valued query g is the most that one record can move it, $\Delta_g = \sup_{x \sim x'} |g(x) - g(x')|$ [6].

The Laplace distribution with location 0 and scale b has density

$$\text{Laplace}(z \mid 0, b) = \frac{1}{2b} e^{-|z|/b}$$

and variance $2b^2$ [6]. For a budget ε', the Laplace mechanism releases $g(x) + z$ with $z \sim \text{Laplace}(0, \Delta_g/\varepsilon')$, and is ε'-differentially private [6]. The noise scale is therefore determined by two quantities, and a smaller ε' or a larger sensitivity requires more noise.

The analysis also uses two further properties of differential privacy. Under basic composition, independent releases with budgets $\varepsilon_1, \ldots, \varepsilon_N$ are together $\left(\sum_{i=1}^{N} \varepsilon_i\right)$-differentially private [6]. Each additional release consumes more of the budget, so a data holder who responds to many queries must accept either weaker privacy or lower accuracy.

Under post-processing, any function of an ε-differentially private output is ε-differentially private [6]. This makes a single reusable release attractive, since analysts may compute any number of quantities from it without weakening the guarantee.

2.3 The private release

The private release scheme studied in this report has four steps. Particles θ_i are drawn from a proposal q that does not depend on the data. Each particle is weighted under the confidential dataset x. Independent zero-mean noise is added to each weight. Finally the particle and noisy-weight pairs are published, and analysts compute estimates of any posterior quantity from them without returning to the data.

Example 1 (continued)**.** The trial publishes N particles, each a candidate value of the five parameters, together with their noisy weights, once. The researchers compute all four quantities above from these pairs, and by post-processing none of them costs the patients any further privacy. With $q = p$ each weight is the likelihood of the trial's records at its particle, so the weights carry the information about the patients and the particles do not. Only the weights therefore require noise.

The Laplace mechanism applies directly to each weight, but zero-mean noise can push a small weight below zero. Standard importance sampling assumes non-negative weights [11], so the scheme relies on the fact that importance sampling extends to real-valued weights [4].

A common alternative is the self-normalised estimator, which divides by the sum of the weights in place of N. This report studies the unnormalised estimator, for which the results hold at every finite N, and leaves self-normalisation to Section 5.2. Diagnostics such as effective sample size and resampling [11] assess the particles and not the estimator, so they are out of scope.

3 Theory of the noisy estimator

This section presents the four results of the report. Propositions 1 and 2 hold for any zero-mean noise that is independent of the particles. Proposition 3 fixes the noise that differential privacy requires, and Proposition 4 puts that noise into Proposition 2.

3.1 Setup and assumptions

Suppose we draw $\theta_1, \ldots, \theta_N \sim q$ i.i.d., for some probability distribution q on the parameter space Θ, the set of values that θ can take. We call q the proposal. The test function $f : \Theta \to \mathbb{R}$ is chosen by the analyst after the release, for example one of the choices in Example 1. The weight function $w : \Theta \times \mathcal{X} \to \mathbb{R}^+$ depends on the observed data $x \in \mathcal{X}$, where $\mathcal{X}$ is the set of possible datasets, and x is fixed throughout. The true weights are $w_i = w(\theta_i, x)$. A noise sample z_i, i.i.d. and independent of the θ_i, is added to each one to form the noisy weights $\hat{w}_i := w_i + z_i$. The two estimators of the target F are

$$\bar{F} = \frac{1}{N} \sum_{i=1}^{N} w_i f(\theta_i) \tag{6}$$

$$\hat{F} = \frac{1}{N} \sum_{i=1}^{N} \hat{w}_i f(\theta_i) \tag{7}$$

The following assumptions are used in the proofs that follow.

1. $\mathrm{E}(\bar{F}) = F$.
2. $\mathrm{var}(\bar{F}) = K_1 < \infty$.
3. $\mathrm{E}(z_i) = 0$ and $\mathrm{var}(z_i) = K_2 < \infty$.

4. $\mathrm{E}(f(\theta_i)) = K_3 < \infty$.
5. $\mathrm{var}(f(\theta_i)) = K_4 < \infty$.

Here F is the integral (3). In our setting, Assumption 1 holds for any valid proposal (Section 2.1), and Assumption 2 states that the noiseless estimator has finite variance. The remaining assumptions require zero-mean noise with finite variance and a test function whose mean and variance under q are finite. Each assumption is cited at the step of a proof that uses it.

3.2 The bias is zero

The bias of an estimator is the difference between its expectation and the target:

$$\mathrm{bias}(\hat{F}) = \mathrm{E}(\hat{F}) - F$$

Proposition 1. *The noisy estimator $\hat{F}$ is unbiased, i.e.,* $\mathrm{bias}(\hat{F}) = 0$.

Proof. We are required to show that $\mathrm{E}(\hat{F}) = F$.

$$\begin{aligned}
\mathrm{E}(\hat{F}) &= \mathrm{E}\left(\frac{1}{N}\sum_{i=1}^{N}(w_i + z_i)f(\theta_i)\right) \\
&= \frac{1}{N}\sum_{i=1}^{N}\left[\mathrm{E}(w_i f(\theta_i)) + \mathrm{E}(z_i f(\theta_i))\right] && \text{by linearity of expectation} \\
&= \frac{1}{N}\sum_{i=1}^{N}\left[\mathrm{E}(w_i f(\theta_i)) + \mathrm{E}(z_i)\ \mathrm{E}(f(\theta_i))\right] && z_i \text{ is independent of } \theta_i \\
&= \frac{1}{N}\sum_{i=1}^{N}\mathrm{E}(w_i f(\theta_i)) && \mathrm{E}(z_i) = 0 \text{ (Assumption 3)}, \\
&&& K_3 \text{ finite (Assumption 4)} \\
&= \mathrm{E}\left(\frac{1}{N}\sum_{i=1}^{N} w_i f(\theta_i)\right) && \text{by linearity of expectation} \\
&= \mathrm{E}(\bar{F}) = F && \text{by Assumption 1}
\end{aligned}$$

□

Hence the noise does not change the estimate on average, at any N and for any test function, including test functions chosen by the analyst after

the release. The noise must also be independent of the particle, because the third line of the proof writes $\mathrm{E}(z_i f(\theta_i))$ as the product $\mathrm{E}(z_i)\,\mathrm{E}(f(\theta_i))$. Without independence this step can fail, and the estimator can be biased even when the noise has mean zero.

Unbiasedness is a statement about the average over many releases. A single release can still be far from F, and the typical distance is measured by the variance in Proposition 2. The result is closely related to random-weight importance sampling, in which the exact weight is replaced by a random weight whose conditional expectation given the particle is proportional to it [4]. Here $\mathrm{E}(\hat{w}_i \mid \theta_i) = w_i$. The setting of this report does not reduce exactly to that framework, but the proof follows the same argument.

3.3 The variance

Proposition 2. *The variance of $\hat{F}$ satisfies:*

$$\mathrm{var}(\hat{F}) = \mathrm{var}(\bar{F}) + \frac{1}{N}\left(K_4 + K_3^2\right) K_2 \tag{8}$$

Proof. Let $g_i = \hat{w}_i f(\theta_i) = w_i f(\theta_i) + z_i f(\theta_i)$. By the law of total variance [1], conditioning on θ_i,

$$\mathrm{var}(g_i) = \mathrm{E}\left(\mathrm{var}(g_i \mid \theta_i)\right) + \mathrm{var}\left(\mathrm{E}(g_i \mid \theta_i)\right) \tag{9}$$

With θ_i fixed, $w_i f(\theta_i)$ and $f(\theta_i)$ are constants, so

$$\begin{aligned}
\mathrm{E}(g_i \mid \theta_i) &= \mathrm{E}\left(w_i f(\theta_i) + z_i f(\theta_i) \mid \theta_i\right) \\
&= w_i f(\theta_i) + f(\theta_i)\ \mathrm{E}(z_i \mid \theta_i) && \text{constants come out once } \theta_i \text{ is fixed, and linearity of expectation} \\
&= w_i f(\theta_i) + f(\theta_i)\ \mathrm{E}(z_i) && z_i \text{ is independent of } \theta_i \\
&= w_i f(\theta_i) && \mathrm{E}(z_i) = 0 \text{ by Assumption 3}
\end{aligned}$$

$$\begin{aligned}
\mathrm{var}(g_i \mid \theta_i) &= \mathrm{var}\left(w_i f(\theta_i) + z_i f(\theta_i) \mid \theta_i\right) \\
&= \mathrm{var}\left(z_i f(\theta_i) \mid \theta_i\right) && w_i f(\theta_i) \text{ is a constant} \\
&= f(\theta_i)^2\ \mathrm{var}(z_i \mid \theta_i) && \text{constant comes out squared} \\
&= f(\theta_i)^2\ \mathrm{var}(z_i) && z_i \text{ is independent of } \theta_i \\
&= f(\theta_i)^2 K_2 && \mathrm{var}(z_i) = K_2 \text{ by Assumption 3}
\end{aligned}$$

For any random variable X, $\mathrm{var}(X) = \mathrm{E}(X^2) - \mathrm{E}(X)^2$, and rearranging gives $\mathrm{E}(X^2) = \mathrm{var}(X) + \mathrm{E}(X)^2$. Applying this with $X = f(\theta_i)$ and Assumptions 4 and 5,

$$\mathrm{E}\left(f(\theta_i)^2\right) = \mathrm{var}(f(\theta_i)) + \mathrm{E}(f(\theta_i))^2 = K_4 + K_3^2$$

Hence, returning to (9),

$$\begin{aligned}
\mathrm{var}(g_i) &= \mathrm{E}\left(\mathrm{var}(g_i \mid \theta_i)\right) + \mathrm{var}\left(\mathrm{E}(g_i \mid \theta_i)\right) \\
&= K_2\ \mathrm{E}\left(f(\theta_i)^2\right) + \mathrm{var}\left(w_i f(\theta_i)\right) \\
&= K_2\left(\mathrm{var}(f(\theta_i)) + \mathrm{E}(f(\theta_i))^2\right) + \mathrm{var}\left(w_i f(\theta_i)\right) \\
&= \left(K_4 + K_3^2\right) K_2 + \mathrm{var}\left(w_i f(\theta_i)\right)
\end{aligned}$$

The g_i are i.i.d., so variances of the sum add:

$$\begin{aligned}
\mathrm{var}(\hat{F}) &= \mathrm{var}\left(\frac{1}{N}\sum_{i=1}^{N} g_i\right) \\
&= \frac{1}{N^2}\sum_{i=1}^{N} \mathrm{var}(g_i) \\
&= \frac{1}{N^2}\sum_{i=1}^{N} \left[\left(K_4 + K_3^2\right) K_2 + \mathrm{var}\left(w_i f(\theta_i)\right)\right] \\
&= \frac{1}{N}\left(K_4 + K_3^2\right) K_2 + \frac{1}{N^2}\sum_{i=1}^{N} \mathrm{var}\left(w_i f(\theta_i)\right) \\
&= \frac{1}{N}\left(K_4 + K_3^2\right) K_2 + \mathrm{var}(\bar{F})
\end{aligned}$$

The last term is finite by Assumption 2. □

The noise contributes one additional term and leaves the Monte Carlo term $\mathrm{var}(\bar{F})$ unchanged. The extra term is the noise variance K_2 times $\mathrm{E}(f(\theta)^2) = K_4 + K_3^2$, divided by N. The uncentred second moment of f appears because the noise multiplies $f(\theta_i)$ itself, not its distance from the mean. Consequently, a test function that takes values far from zero incurs a larger noise term, even if its variance is small. For $f(\theta) = \theta$ under a $\mathrm{N}(0, 1)$ proposal, $K_3 = 0$ and $K_4 = 1$, so the noise term reduces to K_2/N.

Balancing privacy against utility, as set out in Section 1, requires a measure of utility. This report measures utility by the mean squared error of

the estimate, which decomposes into variance plus squared bias [1],

$$\mathrm{MSE}(\hat{F}) = \mathrm{E}\left((\hat{F} - F)^2\right) = \mathrm{var}(\hat{F}) + \left(\mathrm{E}(\hat{F}) - F\right)^2. \tag{10}$$

Proposition 1 makes the second term zero, so $\mathrm{MSE}(\hat{F}) = \mathrm{var}(\hat{F})$ and every variance of $\hat{F}$ below is also its mean squared error.

If K_2 were fixed, both terms would decrease like $1/N$, and the noisy estimator would converge at the same rate as the noiseless one, with a larger constant. Chopin and Papaspiliopoulos [4] show that random weights always increase the asymptotic variance of the self-normalised estimator, and Proposition 2 gives the increase for the unnormalised estimator at every finite N. The next section shows that privacy does not leave K_2 fixed.

3.4 The noise needed for differential privacy

As in Section 2.2, neighbouring datasets $x \sim x'$ differ by replacing one record.

Proposition 3. *Suppose $\theta_i \sim q$ i.i.d. and $w_i = w(\theta_i, x)$ for $i \in \{1, \ldots, N\}$, where $w(\theta, x)$ has sensitivity*

$$\Delta_w = \sup_{\theta} \sup_{x \sim x'} |w(\theta, x) - w(\theta, x')| < \infty \tag{11}$$

Releasing $\{(\theta_i, \hat{w}_i)\}_{i=1}^N$, where $\hat{w}_i = w_i + z_i$, is ε-differentially private if, i.i.d. and independent of the θ_i,

$$z_i \sim \mathrm{Laplace}\left(0, \frac{N\Delta_w}{\varepsilon}\right) \tag{12}$$

Proof. The proposal q does not depend on x, so releasing the θ_i costs no privacy. Allocate an equal share ε' of the budget to each weight, so $N\varepsilon' = \varepsilon$. Then $\hat{w}_i$ is ε'-differentially private by the Laplace mechanism, and the N independent releases together are ε-differentially private by basic composition. Writing b for the Laplace scale, this noise has mean zero and finite variance, so Assumption 3 holds, with scale and variance

$$\begin{aligned} b &= \frac{\Delta_w}{\varepsilon'} = \frac{N\Delta_w}{\varepsilon} && \text{Laplace mechanism, then } \varepsilon' = \tfrac{\varepsilon}{N} \\ K_2 &= 2b^2 = \frac{2N^2\Delta_w^2}{\varepsilon^2} && \text{variance of Laplace}(0, b) \end{aligned} \tag{13}$$

A direct proof from the densities, without the cited theorems, is in Appendix A. □

Let the dataset $x = (x_1, \ldots, x_n)$ consist of n records that are independent given θ. With $q = p$ the weight is the likelihood of these records,

$$w(\theta, x) = \prod_{j=1}^{n} L(x_j \mid \theta).$$

Let $L_{\max} = \sup_{\theta, x_1} L(x_1 \mid \theta)$ be the largest value that one record's likelihood can take. Every factor lies in $[0, L_{\max}]$, so $0 \le w(\theta, x) \le L_{\max}^n$. Replacing one record changes one factor by at most $L_{\max}$ and leaves $n-1$ factors that are each at most $L_{\max}$, so the sensitivity is at most $L_{\max}^n$. This report takes $\Delta_w = L_{\max}^n$, since an upper bound on the sensitivity suffices for the Laplace mechanism.

The scale depends on the choices made in the proof, mainly the equal division of the budget and basic composition. It is sufficient under those choices, and other mechanisms may require less noise. Appendix A shows that any larger scale also gives ε-differential privacy.

Consequently, the noise on each weight grows with N: doubling the number of particles halves the budget of each weight, doubles b and quadruples K_2.

3.5 The variance under DP noise, and the best N

Proposition 2 holds at each fixed N. Here $K_2 = \mathrm{var}(z_i)$ is set by the privacy calibration of Proposition 3, with $K_2 = 2N^2\Delta_w^2/\varepsilon^2$, to maintain the privacy budget, so it depends on N.

Proposition 4. *Under the calibration of Proposition 3,*

$$\mathrm{var}(\hat{F}) = V(N) = \frac{A}{N} + BN, \tag{14}$$

where

$$A = \mathrm{var}\left(w(\theta, x) f(\theta)\right), \qquad B = \frac{2\Delta_w^2 \left(K_4 + K_3^2\right)}{\varepsilon^2}$$

Treating $N > 0$ as a real number, V is minimised at N_c, with the smallest variance $V(N_c)$:

$$N_c = \sqrt{\frac{A}{B}}, \qquad V(N_c) = 2\sqrt{AB}$$

Proof. From Proposition 2,

$$\operatorname{var}(\hat{F}) = \operatorname{var}(\bar{F}) + \frac{1}{N}\left(K_4 + K_3^2\right) K_2$$

where $K_4 = \operatorname{var}(f(\theta))$, $K_3^2 = \mathrm{E}(f(\theta))^2$ and $K_2 = \operatorname{var}(z_i)$, so $K_4 + K_3^2 = \mathrm{E}\left(f(\theta)^2\right)$ is the uncentred second moment of f.

To find the optimal N, the variance must be written explicitly as a function of N. In Proposition 2 this dependence is partly hidden, because $\operatorname{var}(\bar{F})$ and K_2 also depend on N.

First we consider the per-particle variance with no additional noise, $A = \operatorname{var}\left(w(\theta, x) f(\theta)\right)$. It does not depend on N.

$$\begin{aligned}
\operatorname{var}(\bar{F}) &= \operatorname{var}\left(\frac{1}{N}\sum_{i=1}^{N} w_i f(\theta_i)\right) \\
&= \frac{1}{N^2}\sum_{i=1}^{N} \operatorname{var}\left(w_i f(\theta_i)\right) \qquad \theta_i \text{ i.i.d., } \tfrac{1}{N} \text{ comes out squared} \\
&= \frac{1}{N^2} \cdot NA \\
&= \frac{A}{N}
\end{aligned}$$

Now we consider the variance from the privatising noise. Recall that $K_2 = \operatorname{var}(z_i)$ is the variance of this noise, and that the proof of Proposition 3 sets it to $K_2 = 2N^2\Delta_w^2/\varepsilon^2$. Substituting into the noise term of Proposition 2,

$$\begin{aligned}
\frac{1}{N}\left(K_4 + K_3^2\right) K_2 &= \frac{1}{N}\left(K_4 + K_3^2\right) \frac{2N^2\Delta_w^2}{\varepsilon^2} \\
&= \frac{2\Delta_w^2\left(K_4 + K_3^2\right)}{\varepsilon^2} N \\
&= BN
\end{aligned}$$

That is, B is the uncentred second moment of f multiplied by K_2/N^2, so the noise contributes BN.

Therefore

$$V(N) = \operatorname{var}(\hat{F}) = \frac{A}{N} + BN$$

The first term decreases with N and the second increases.

Treating $N > 0$ as a real number, the minimum is found as follows.

$$\begin{aligned} V'(N) &= -\frac{A}{N^2} + B \\ 0 &= -\frac{A}{N_c^2} + B \\ N_c &= \sqrt{\frac{A}{B}} \\ V''(N) &= \frac{2A}{N^3} > 0 \text{ for } N > 0 \end{aligned}$$

Hence V is convex for $N > 0$ and N_c is a minimum.

Substituting N_c into V, with $1/N_c = \sqrt{B/A}$, the smallest variance reachable is

$$\begin{aligned} V(N_c) &= A\sqrt{\frac{B}{A}} + B\sqrt{\frac{A}{B}} \\ &= \sqrt{AB} + \sqrt{AB} \\ &= 2\sqrt{AB} \end{aligned}$$

□

The term A/N is the ordinary Monte Carlo variance of Section 2.1, and it decreases as particles are added. The term BN is the contribution of the privacy noise. Each additional particle is one more weight to protect, so the budget per weight decreases to ε/N, the noise variance per weight grows like N^2, while Proposition 2 divides it only by N. At N_c the two terms are equal, $A/N_c = BN_c = \sqrt{AB}$. Below N_c an additional particle reduces the variance, and above N_c every additional particle increases it.

No choice of N reduces the variance below $2\sqrt{AB}$, so at a fixed ε additional computation cannot drive it to zero. Since $\hat{F}$ is unbiased, Chebyshev's inequality [1] gives, for any $c > 0$,

$$\Pr(|\hat{F} - F| > c) \leq \frac{\operatorname{var}(\hat{F})}{c^2}.$$

If the variance tends to zero, this bound establishes consistency. Under basic composition $\operatorname{var}(\hat{F}) \geq 2\sqrt{AB}$ for every N, so the bound does not tend to zero and consistency cannot be established by this argument.

The lower bound $2\sqrt{AB}$ is large when A or B is large. Since B grows like Δ_w^2/ε^2, the bound is proportional to Δ_w/ε, so a larger sensitivity raises

the bound in proportion and halving the budget doubles the bound. Even when the bound is small, not every N is acceptable, because $V(N)$ increases without bound once N exceeds N_c.

4 Simulation

This section checks Proposition 4 numerically on a small normal model, and then measures how much a release reveals about one record. The code for both experiments is available at `https://github.com/wakhee/math-7000-final-submission`.

4.1 Model and settings

The dataset has $n = 10$ records. Given θ, the records are independent with $x_j \sim \mathrm{N}(\theta, 1)$, and the data are generated with $\theta = 1$. The prior is $\mathrm{N}(0, 1)$ and, as in Section 2.1, the proposal is the prior, so the weight is the likelihood,

$$w(\theta, x) = \prod_{j=1}^{n} \frac{1}{\sqrt{2\pi}} \exp\left(-\tfrac{1}{2}(x_j - \theta)^2\right).$$

One record's likelihood is largest when $x_j = \theta$, so $L_{\max} = 1/\sqrt{2\pi}$ and $\Delta_w = L_{\max}^n = (2\pi)^{-5} \approx 1.02 \times 10^{-4}$. The noise on each weight is $\mathrm{Laplace}(0, N\Delta_w/\varepsilon)$, as in (12). The test function is $f(\theta) = \theta$, for which $K_3 = 0$ and $K_4 = 1$ (Section 3.3), so $B = 2\Delta_w^2/\varepsilon^2$.

4.2 Variance of the estimator

One dataset is generated and held fixed, as in Section 3.1. For each $N \in \{1, 3, 10, 30, 100, 300, 1000\}$, 1000 independent releases are simulated, each with new particles and new noise, and the sample variance of the 1000 values of $\hat{F}$ is compared with $V(N) = A/N + BN$. The constant A is estimated from 10^6 draws from the prior, which gives $A \approx 2.3 \times 10^{-13}$.

Figure 1 shows the result at two budgets. At $\varepsilon = 0.1$, which is in the strong-privacy range of Section 2.2, $N_c \approx 3.3 \times 10^{-4}$ is below one. The noise term dominates at every $N \geq 1$, the variance increases with N, and

a single particle gives the smallest variance. The worst-case sensitivity is the reason: the standard deviation of one term $w(\theta, x)f(\theta)$ is $\sqrt{A} \approx 4.8 \times 10^{-7}$, about 200 times smaller than Δ_w. To make the minimum of V visible, the budget has to be raised to $\varepsilon = 10^4$, which is far too large to give meaningful privacy. Then $N_c \approx 33$, and the measured variance decreases up to about $N = 30$ and increases afterwards. At both budgets the ratio of the measured variance to $V(N)$ lies between 0.91 and 1.15 at every N.

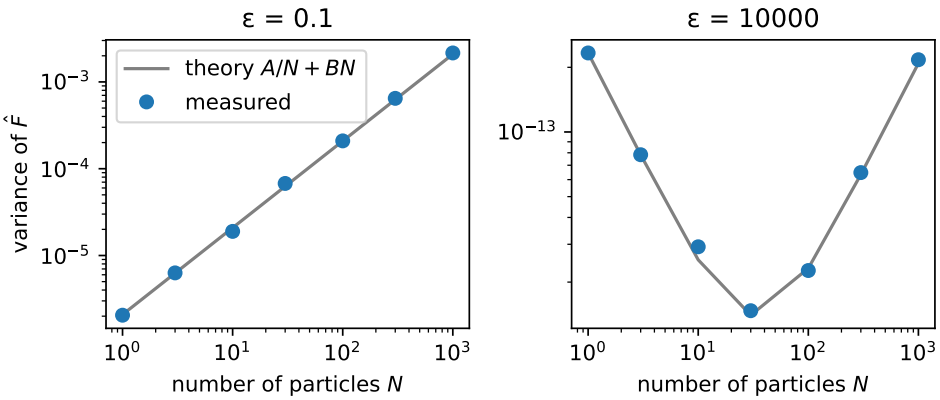


Figure 1: Variance of $\hat{F}$ against the number of particles N for the model of Section 4.1, at $\varepsilon = 0.1$ (left) and $\varepsilon = 10^4$ (right). The dots are sample variances over 1000 simulated releases, and the grey line is $V(N) = A/N + BN$ from Proposition 4. Both axes are logarithmic, and the vertical scales of the two panels differ.

4.3 An attack on one record

The second experiment measures what a release reveals about one record. An adversary is assumed to know the model, the released pairs $(\theta_i, \hat{w}_i)$ and nine of the ten records, and tries to recover the remaining record x_2. For a candidate value u, let $x^{(u)}$ be the dataset with x_2 replaced by u. Under Laplace noise with scale b, the log-likelihood of the released weights is $-\sum_i |\hat{w}_i - w(\theta_i, x^{(u)})|/b$ up to a constant, so the maximum-likelihood guess is

$$\hat{x}_2 = \arg\min_u \sum_{i=1}^{N} \left| \hat{w}_i - w(\theta_i, x^{(u)}) \right| ,$$

which does not depend on b. The minimum is found by evaluating the sum at 101 equally spaced values of u in $[-5, 5]$ and refining locally from the best of them.

The release has $N = 1000$ particles, and 15 budgets are used, equally spaced on a logarithmic scale from 10^{-2} to 10^{9}. Each of 1000 repetitions generates a new dataset and new particles, adds fresh noise at each budget, and records the squared error $(\hat{x}_2 - x_2)^2$. The reference is an adversary who ignores the release and guesses the mean of the nine known records. The error of that guess is x_2 minus the mean of nine other records, which has variance $1 + 1/9$, so its mean squared error is about 1.11.

Figure 2 shows the mean squared error of the attack at each budget. For budgets up to about 5×10^2 it lies between 12.7 and 17.6, more than ten times the reference, so over this range the attack does worse than ignoring the release. The range includes every budget classed as strong or moderate privacy in Section 2.2. The values in this range differ by more than their error bars, and the cause of this variation has not been identified. The error then decreases, to 5.2 at $\varepsilon \approx 1.9 \times 10^4$ and 2.1 at $\varepsilon \approx 1.2 \times 10^5$, and it first falls below the reference at $\varepsilon \approx 7.2 \times 10^5$, where it is 0.27.

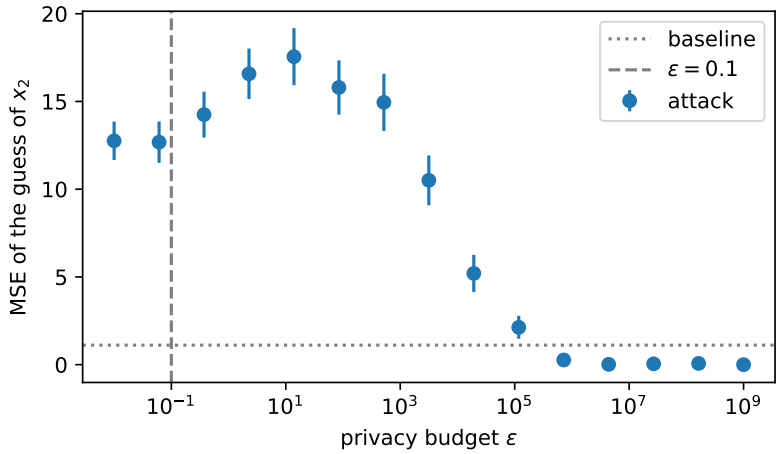


Figure 2: Mean squared error of the adversary's guess of x_2 against the privacy budget ε (logarithmic axis), with $N = 1000$ particles. Each dot is the mean over 1000 repetitions, and the bars are ± 1.96 standard errors. The dotted line is the reference value $1 + 1/9$, the error of guessing the mean of the nine known records. The dashed line marks $\varepsilon = 0.1$.

For comparison, setting $A/N = BN$ with $B = 2\Delta_w^2/\varepsilon^2$ shows that at

$N = 1000$ the two terms of $V(N)$ are equal when $\varepsilon = N\Delta_w\sqrt{2/A}$, which is about 3.0×10^5 for the dataset of Section 4.2. In this example the attack therefore begins to beat the reference at budgets of the same order as those at which the noise stops dominating the variance of the estimate. The experiment uses one model and one attack. A failed attack does not establish privacy, which is guaranteed by Proposition 3, and the comparison only illustrates the trade-off between privacy and utility under the worst-case sensitivity.

5 Discussion

Proposition 2 shows that when the noise variance K_2 is held fixed while N grows, the mean squared error of the unnormalised estimator decreases like $1/N$, so by Chebyshev's inequality the estimator is consistent. Proposition 4 instead lets the noise scale grow with N, as a fixed budget requires. The $1/N$ decay then stops and the lower bound $2\sqrt{AB}$ appears. The two results describe different settings, and a data holder with a fixed ε is in the second.

The sign of a noisy weight plays no part in Propositions 1 and 2, whose proofs use only the mean and variance of the noise and its independence from the particles. This agrees with the observation of Chopin and Papaspiliopoulos [4] that random weights need not be non-negative. Accuracy is lost through the magnitude of the noise, which Proposition 4 quantifies.

5.1 Limitations

The estimator studied here targets $F = Z\ \mathrm{E}(f(\theta) \mid x)$, not the posterior expectation itself. An analyst who wants $\mathrm{E}(f(\theta) \mid x)$ has to divide by an estimate of Z, which is the self-normalised estimator, and none of the results above cover this situation.

The sensitivity $\Delta_w = L^n_{\max}$ is set by the worst dataset and the worst parameter value, not by the data at hand. Weights can be far smaller than Δ_w, while the noise is calibrated to Δ_w, so it can be large compared with every weight in the release. A tighter sensitivity would reduce B, which would increase N_c and lower the bound $2\sqrt{AB}$.

The analysis uses pure ε-differential privacy, the Laplace mechanism and basic composition. Basic composition adds the budgets of the N releases, and the linear growth of the noise term in N is a consequence of this choice. Other composition results can give a smaller total budget for many releases [6].

The proposal is taken to be the prior. A proposal closer to the posterior could lower A, but it must not depend on x, or releasing the particles would itself cost privacy. Finiteness of A, K_3 and K_4 is assumed here. It holds whenever the weights are bounded and f has a finite second moment under q. The simulation of Section 4 covers one model, one test function and one attack.

5.2 Future work

A natural extension is the self-normalised estimator, where noise enters both the numerator and the denominator of the ratio. A denominator that can take either sign can be close to zero, so an initial question is whether the moments of the estimator exist at finite N.

Different privacy accounting may change the shape of (14). One candidate is advanced composition, which uses approximate differential privacy: an algorithm is (ε, δ)-differentially private if (5) holds with an additive term δ on the right-hand side [6]. Under advanced composition, N releases that are each ε'-differentially private are together $(\varepsilon_{\mathrm{tot}}, \delta')$-differentially private with

$$\varepsilon_{\mathrm{tot}} = \sqrt{2N \ln(1/\delta')}\, \varepsilon' + N\varepsilon'(e^{\varepsilon'} - 1)$$

[6]. If the first term dominates, the scale needed per weight grows like $\sqrt{N}$ rather than N, at the cost of a small failure probability δ'. A heuristic substitution into (8) then suggests that the noise term stays roughly constant in N rather than growing linearly. The variance would still not tend to zero, but adding particles might no longer increase it. This has not been established here, and a careful analysis would need to account for the second term and for δ'. Concentrated differential privacy [3] and Rényi differential privacy [10], a relaxation based on the Rényi divergence, offer further accounting methods for composed mechanisms, and it may be fruitful to examine whether either yields a more favourable trade-off.

Another possible direction is local sensitivity, which measures how much one record can change the weight at the observed dataset rather than in

the worst case. It remains to be checked whether an approach of this kind can give a valid guarantee with less noise, which might ease the limitation on Δ_w described above.

The simulation of Section 4 could be extended to other models and test functions, and to the self-normalised estimator. An empirical audit [9] might then be used to estimate a lower bound on the privacy budget that a release spends in practice.

6 Conclusion

This report studied the unnormalised importance-sampling estimator when zero-mean noise is added to the weights before release. The noisy estimator is unbiased at every N, and its variance is that of the noise-free estimator plus a term proportional to the noise variance. This agrees with the observation of Chopin and Papaspiliopoulos [4] that random weights need not be non-negative for importance sampling to remain valid.

Calibrating Laplace noise for ε-differential privacy under basic composition makes the noise variance per weight grow like N^2, and the mean squared error of the estimator, which equals its variance, becomes $A/N + BN$. It is bounded below by $2\sqrt{AB}$ for every N, so at a fixed budget additional computation cannot make the estimator arbitrarily accurate, and the Chebyshev argument for consistency cannot be completed. With the noise variance held fixed the estimator is consistent, and this report shows what changes once a fixed budget ties the noise to N.

In a simulation on a normal model with ten records, the measured variance followed $A/N + BN$, and an attack on one record did better than a simple reference guess only at budgets above about 10^5.

A Direct proof of Proposition 3

Proposition 3 relied on two cited theorems. This appendix proves the privacy claim directly from the densities, and shows that any scale $b \geq \dfrac{N\Delta_w}{\varepsilon}$ suffices.

A release with density p_x under dataset x is ε-differentially private if, for

all neighbours $x \sim x'$ and every output o,

$$p_x(o) \leq e^{\varepsilon} p_{x'}(o)$$

Integrating both sides over any set S of outputs gives (5).

The Laplace density with location μ and scale b is

$$\text{Laplace}(z \mid \mu, b) = \frac{1}{2b} e^{-|z-\mu|/b}$$

The θ_i are drawn from the proposal q, so each has density $q(\theta_i)$. Given θ_i, the released weight $\hat{w}_i = w(\theta_i, x) + z_i$ has density

$$\text{Laplace}\left(\hat{w}_i \mid w(\theta_i, x), b\right)$$

Fix one released pair $(\theta, \hat{w})$. Write the true weight under each dataset as

$$c = w(\theta, x), \qquad c' = w(\theta, x'), \qquad \text{so } |c - c'| \leq \Delta_w$$

The ratio of the two Laplace densities at $\hat{w}$ is

$$\begin{aligned} \frac{\text{Laplace}(\hat{w} \mid c, b)}{\text{Laplace}(\hat{w} \mid c', b)} &= \frac{\frac{1}{2b} e^{-|\hat{w}-c|/b}}{\frac{1}{2b} e^{-|\hat{w}-c'|/b}} \\ &= e^{(-|\hat{w}-c|+|\hat{w}-c'|)/b} \\ &= e^{(|\hat{w}-c'|-|\hat{w}-c|)/b} \end{aligned}$$

Now bound the exponent. By the triangle inequality, $|\hat{w} - c'| \leq |\hat{w} - c| + |c - c'|$, so

$$|\hat{w} - c'| - |\hat{w} - c| \leq |c - c'| \leq \Delta_w$$

Dividing by b and applying e^u, which is increasing in u, the ratio above satisfies, for one pair,

$$\frac{\text{Laplace}(\hat{w} \mid c, b)}{\text{Laplace}(\hat{w} \mid c', b)} \leq e^{\Delta_w / b}$$

Hence one pair is $\frac{\Delta_w}{b}$-differentially private, that is, it uses at most $\frac{\Delta_w}{b}$ of the budget.

The pairs are independent, so the density of the whole release at an output $o = (\theta_1, \hat{w}_1, \ldots, \theta_N, \hat{w}_N)$ is a product with one factor per pair:

$$p_x(o) = \prod_{i=1}^{N} q(\theta_i)\,\mathrm{Laplace}\,(\hat{w}_i \mid w(\theta_i, x), b)$$

Each factor is the density of θ_i times the density of $\hat{w}_i$ given θ_i. Only the second depends on the dataset x. Hence

$$\begin{aligned}
\frac{p_x(o)}{p_{x'}(o)} &= \prod_{i=1}^{N} \frac{q(\theta_i)\,\mathrm{Laplace}\,(\hat{w}_i \mid w(\theta_i, x), b)}{q(\theta_i)\,\mathrm{Laplace}\,(\hat{w}_i \mid w(\theta_i, x'), b)} \\
&= \prod_{i=1}^{N} \frac{\mathrm{Laplace}\,(\hat{w}_i \mid w(\theta_i, x), b)}{\mathrm{Laplace}\,(\hat{w}_i \mid w(\theta_i, x'), b)} && q(\theta_i) \text{ does not depend on } x, \text{ so cancels} \\
&\leq \prod_{i=1}^{N} e^{\Delta_w / b} \\
&= e^{N\Delta_w / b}
\end{aligned}$$

This is at most e^{ε} exactly when $\dfrac{N\Delta_w}{b} \leq \varepsilon$, that is $b \geq \dfrac{N\Delta_w}{\varepsilon}$.

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